

```
-- patient table
create table patient (
    patientid int auto_increment primary key,
    firstname varchar(50),
    lastname varchar(50),
    dateofbirth date,
    gender varchar(10),
    contactinfo varchar(100)
);
```

```
-- doctor table
create table doctor (
    doctorid int auto_increment primary key,
    firstname varchar(50),
    lastname varchar(50),
    specialty varchar(50),
    phone varchar(15),
    email varchar(100)
);
```

```
-- medicalrecord table
create table medicalrecord (
    recordid int auto_increment primary key,
    patientid int,
    diagnosis varchar(100),
    treatment varchar(100),
    recorddate date,
    notes varchar(255),

    foreign key (patientid)
    references patient(patientid)
);
```

```
-- billing table
create table billing (
    billingid int auto_increment primary key,
    patientid int,
    amount decimal(10,2),
```

```
    billingdate date,  
    status varchar(20),  
  
    foreign key (patientid)  
    references patient(patientid)  
);  
  
-- appointment table  
create table appointment (  
    appointmentid int auto_increment primary key,  
    patientid int,  
    doctorid int,  
    appointmentdate date,  
    reason varchar(100),  
    status varchar(20),  
  
    foreign key (patientid)  
    references patient(patientid),  
  
    foreign key (doctorid)  
    references doctor(doctorid)  
);  
  
-- prescription table  
create table prescription (  
    prescriptionid int auto_increment primary key,  
    patientid int,  
    doctorid int,  
    medication varchar(100),  
    dosage varchar(50),  
    startdate date,  
    enddate date,  
  
    foreign key (patientid)  
    references patient(patientid),  
  
    foreign key (doctorid)  
    references doctor(doctorid)
```

);

-- patient data

insert into patient values

(1,'arjun','shrestha','2000-01-10','male','9811111111'),
(2,'aayush','karki','1999-05-20','male','9822222222'),
(3,'bina','adhikari','2001-03-15','female','9833333333'),
(4,'chandra','gautam','1998-07-22','male','9844444444'),
(5,'diya','thapa','2002-11-05','female','9855555555'),
(6,'eric','lama','1997-09-30','male','9866666666'),
(7,'fiza','rana','2001-12-12','female','9877777777'),
(8,'gopal','basnet','1996-04-18','male','9888888888'),
(9,'hema','chaudhary','2003-08-08','female','9899999999'),
(10,'ishwar','joshi','1995-02-25','male','9801010101');

-- doctor data

insert into doctor values

(1,'sanjay','poudel','cardiology','9841000001','sanjay@gmail.com'),
(2,'nisha','shrestha','dermatology','9841000002','nisha@gmail.com'),
(3,'rohit','bhatt','orthopedics','9841000003','rohit@gmail.com'),
(4,'anita','karki','neurology','9841000004','anita@gmail.com'),
(5,'manoj','adhikari','dentistry','9841000005','manoj@gmail.com'),
(6,'priyanka','yadav','gynecology','9841000006','priyanka@gmail.com'),
(7,'alok','rana','ent','9841000007','alok@gmail.com'),
(8,'sunita','shrestha','general medicine','9841000008','sunita@gmail.com'),
(9,'kabir','thapa','pediatrics','9841000009','kabir@gmail.com'),
(10,'meena','chaudhary','psychiatry','9841000010','meena@gmail.com');

-- medicalrecord data

insert into medicalrecord values

(1,1,'hypertension','medicine','2026-01-02','high bp'),
(2,2,'skin allergy','cream','2026-01-05',null),
(3,3,'fracture','surgery','2026-01-08','operation done'),
(4,4,'migraine','painkiller','2026-01-10','follow up'),
(5,5,'fever','paracetamol','2026-01-12',null),
(6,6,'ear infection','antibiotics','2026-01-14','mild infection'),
(7,7,'tooth cavity','surgery','2026-01-16','surgery planned'),
(8,8,'pregnancy','checkup','2026-01-18',null),

```
(9,9,'back pain','therapy','2026-01-20','exercise advised'),  
(10,10,'depression','counselling','2026-01-22',null);
```

-- billing data

insert into billing values

```
(1,1,1500,'2026-01-03','paid'),  
(2,2,800,'2026-01-06','paid'),  
(3,3,2500,'2026-01-09','pending'),  
(4,4,1200,'2026-01-11','paid'),  
(5,5,600,'2026-01-13','paid'),  
(6,6,1750,'2026-01-15','pending'),  
(7,7,2000,'2026-01-17','paid'),  
(8,8,950,'2026-01-19','paid'),  
(9,9,700,'2026-01-21','paid'),  
(10,10,500,'2026-01-24','pending');
```

-- appointment data

insert into appointment values

```
(1,1,1,'2026-01-10','heart checkup','scheduled'),  
(2,2,2,'2026-02-11','skin allergy','completed'),  
(3,3,3,'2026-03-15','bone pain','scheduled'),  
(4,4,4,'2025-12-20','headache','cancelled'),  
(5,5,5,'2026-01-18','tooth pain','scheduled'),  
(6,6,1,'2026-02-22','ear pain','completed'),  
(7,7,2,'2026-04-01','pregnancy checkup','scheduled'),  
(8,8,3,'2026-05-12','back pain','completed'),  
(9,9,1,'2026-01-25','bp follow up','scheduled'),  
(10,10,8,'2026-02-05','mental health','scheduled');
```

-- prescription data

insert into prescription values

```
(1,1,1,'amlodipine','5mg','2026-01-02','2026-02-02'),  
(2,2,2,'cetirizine','10mg','2026-01-05','2026-01-12'),  
(3,3,1,'paracetamol','500mg','2026-01-08','2026-01-15'),  
(4,4,1,'ibuprofen','400mg','2026-01-10','2026-01-17'),  
(5,5,2,'amoxicillin','500mg','2026-01-12','2026-01-19'),  
(6,6,2,'paracetamol','500mg','2026-01-14','2026-01-20'),  
(7,7,3,'vitamin d','1 tablet','2026-01-16','2026-02-16'),
```

```
(8,8,3,'folic acid','1 tablet','2026-01-18','2026-02-18'),  
(9,9,1,'atenolol','50mg','2026-01-20','2026-02-20'),  
(10,10,2,'sertraline','50mg','2026-01-22','2026-02-22');
```

-- question 1

```
select patientid, sum(amount) as total_amount  
from billing  
group by patientid;
```

-- question 2

```
select  
concat(d.firstname, ' ', d.lastname) as doctorname,  
count(a.appointmentid) as appointment_count  
from doctor d  
left join appointment a  
on d.doctorid = a.doctorid  
group by d.doctorid;
```

-- question 3

```
select avg(amount) as average_billing  
from billing;
```

-- question 4

```
select  
max(amount) as maximum_amount,  
min(amount) as minimum_amount  
from billing;
```

-- question 5

```
select  
concat(d.firstname, ' ', d.lastname) as doctorname,  
count(p.prescriptionid) as prescription_count  
from doctor d  
join prescription p  
on d.doctorid = p.doctorid  
group by d.doctorid  
having count(p.prescriptionid) > 2;
```

-- question 6

```
select diagnosis, count(*) as record_count
from medicalrecord
group by diagnosis;
```

-- question 7

```
select
concat(p.firstname, ' ', p.lastname) as patient_name,
m.diagnosis
from patient p
join medicalrecord m
on p.patientid = m.patientid;
```

-- question 8

```
select
a.appointmentdate,
a.reason,
a.status,
concat(p.firstname, ' ', p.lastname) as patient_name,
concat(d.firstname, ' ', d.lastname) as doctor_name
from appointment a
join patient p
on a.patientid = p.patientid
join doctor d
on a.doctorid = d.doctorid;
```

-- question 9

```
select
pr.medication,
concat(p.firstname, ' ', p.lastname) as patient_name,
concat(d.firstname, ' ', d.lastname) as doctor_name
from prescription pr
join patient p
on pr.patientid = p.patientid
join doctor d
on pr.doctorid = d.doctorid;
```

-- question 10

```
select
p.patientid,
concat(p.firstname,' ',p.lastname) as patient_name,
b.amount,
b.status
from patient p
left join billing b
on p.patientid = b.patientid;
```

-- question 11

```
select
concat(d.firstname,' ',d.lastname) as doctor_name,
a.appointmentdate,
a.reason
from doctor d
left join appointment a
on d.doctorid = a.doctorid;
```

-- question 12

```
select
concat(p.firstname,' ',p.lastname) as patient_name,
sum(b.amount) as total_bill
from patient p
join billing b
on p.patientid = b.patientid
group by p.patientid;
```

-- question 13

```
select
a.appointmentdate,
a.reason,
a.status,
p.gender,
p.contactinfo
from appointment a
join patient p
on a.patientid = p.patientid;
```

```
-- question 14
select *
from appointment
where status = 'scheduled'
and appointmentdate > '2026-01-01';
```

```
-- question 15
select *
from patient
where firstname like 'a%';
```

```
-- question 16
select *
from billing
where amount between 500 and 2000;
```

```
-- question 17
select *
from prescription
where medication in ('paracetamol','ibuprofen','amoxicillin');
```

```
-- question 18
select *
from medicalrecord
where treatment like '%surgery%'
or notes is not null;
```

```
-- question 19
alter table patient
add bloodgroup varchar(5);
```

```
update patient set bloodgroup='a+' where patientid=1;
update patient set bloodgroup='b+' where patientid=2;
update patient set bloodgroup='o+' where patientid=3;
update patient set bloodgroup='ab+' where patientid=4;
update patient set bloodgroup='a-' where patientid=5;
update patient set bloodgroup='b-' where patientid=6;
update patient set bloodgroup='o-' where patientid=7;
```



```
update patient set bloodgroup='ab-' where patientid=8;  
update patient set bloodgroup='a+' where patientid=9;  
update patient set bloodgroup='o+' where patientid=10;
```

-- question 20

```
alter table doctor  
modify phone varchar(20);
```